

## 1. Data Structures

### Target Structure

Purpose: Used to store the name, latitude, and longitude of a target.

### TempStorage Structure

Purpose: Temporarily stores target data read from a file, which is then added to the main target array after validation.

## 2. Functions

- ``load_Target_File``: Loads and validates the target file.
- ``validate_Decimal_Format``: Determine whether the decimal places are correct.
- ``addTarget``: Adds a validated target to the main array.
- ``get_valid_option``: Retrieves the user's selected option.
- ``is_Valid_Name``: Checks if the target name is valid.
- ``is_Valid_Coordinate``: Verifies if the coordinates are valid.
- ``get_Valid_Latitude`` and ``get_Valid_Longitude``: Obtain valid latitude and longitude from user input.
- ``get_Valid_Radius``: Retrieves a valid radius value.
- ``is_Target_in``: Determines if a target is within a given range.
- ``plan_Airstrike``: Plans an airstrike, identifying all targets within a specified range.
- ``execute_Airstrike``: Executes an airstrike, simulating the destruction of targets.
- ``show_Current_Targets``: Displays all current targets.
- ``search_Target``: Searches for a specific target.

## 3. Program Interaction

The program interacts with the user through the console, offering options to load a target file, display targets, search for a target, plan an airstrike, and execute an airstrike. Users perform actions by entering their chosen options.

## 4. Target File Validity

A target file is considered valid if:

- Each line contains exactly three space-separated data items: target name, latitude, and longitude.
- The target name consists only of letters and numbers, not exceeding 15 characters in length.
- Latitude and longitude are valid floating-point numbers within the specified range (latitude: 0-100, longitude: 0-100), with no more than 12 decimal places, and do not include any symbols (except for the decimal point).

## 5. Criteria for Identifying Valid Targets

Valid targets must meet the following criteria:

- The name is valid (letters and numbers, length  $\leq 15$ ).
- Latitude and longitude are between 0 and 100, with no more than 15 decimal places, and do not include any symbols (except for the decimal point and a positive sign).

## 6. Minimum Distance Between Targets

The program does not explicitly set a minimum distance between targets. However, the ``is_Target_in`` function can calculate and determine if a target is within a given radius, which can be used to simulate the distance between targets.

## 7. Memory Leak Testing

To test the program for memory leaks, the following methods can be used:

- **Valgrind**: Run the program with the Valgrind tool to detect memory leaks.
- **Dynamic Memory Management Check**: Ensure that all memory allocated with ``malloc`` is freed with ``free`` when no longer needed.
- **Code Review**: Examine the memory allocation and deallocation logic in the code to ensure nothing is missed.

## 9. Conclusion

This program implements target management, airstrike planning, and execution through structured data structures and functions. Detailed input validation ensures the accuracy and robustness of the program. Memory leak testing and prevention measures guarantee the long-term stability and performance of the program.